

MariaDB [CaseStudy2]> show tables;

```
+-----+
| Tables_in_casestudy2 |
+-----+
| department           |
| employee              |
| project               |
| workson               |
+-----+
```

MariaDB [CaseStudy2]> ALTER TABLE Employee ADD CONSTRAINT CHECK (sex IN ('M','F'));

MariaDB [CaseStudy2]> ALTER TABLE Employee ADD CONSTRAINT CHECK (position IN ('Manager', 'Team Leader', 'Analyst', 'Software Developer'));

MariaDB [CaseStudy2]> ALTER TABLE WorksOn MODIFY hoursWorked INT CHECK (hoursWorked >= 0 AND hoursWorked <= 40);

CREATE VIEW FemaleManagerProjects AS SELECT p.projNo, p.projName, p.deptNo FROM Project p JOIN Department d ON p.deptNo = d.deptNo JOIN Employee e ON d.mgrEmpNo = e.empNo WHERE e.sex = 'F' ORDER BY p.projNo;

MariaDB [CaseStudy2]> select * from FemaleManagerProjects;

```
+-----+-----+-----+
| projNo | projName          | deptNo |
+-----+-----+-----+
| P03    | City Master Plan  | D02    |
| P04    | Land Use Survey   | D02    |
+-----+-----+-----+
```

Section 7.26

MariaDB [CaseStudy2]> create view EmployeesProjectDuration as select emp.empNo, emp.fName, emp.lName, proj.projName, wn.hoursWorked from workson wn join Employee emp on wn.empNo = emp.empNo join Project proj on wn.projNo = proj.projNo;

MariaDB [CaseStudy2]> select * from EmployeesProjectDuration;

```
+-----+-----+-----+-----+-----+
| empNo | fName | lName | projName          | hoursWorked |
+-----+-----+-----+-----+-----+
| E001  | John  | Mensah | Network Upgrade   | 7           |
| E001  | John  | Mensah | ERP Implementation | 6           |
| E002  | Ama  | Owusu  | Network Upgrade   | 8           |
| E002  | Ama  | Owusu  | ERP Implementation | 6           |
| E003  | Kwame | Asante | ERP Implementation | 7           |
| E004  | Efua  | Boateng | City Master Plan  | 8           |
| E004  | Efua  | Boateng | Land Use Survey    | 7           |
| E005  | Yaw   | Darko  | City Master Plan  | 6           |
| E005  | Yaw   | Darko  | Land Use Survey    | 4           |
| E006  | Akosua | Frimpong | Land Use Survey    | 8           |
| E007  | Kofi  | Appiah | Budget Forecast 2026 | 8           |
| E007  | Kofi  | Appiah | Audit Review       | 8           |
| E008  | Adwoa | Sarpong | Budget Forecast 2026 | 7           |
+-----+-----+-----+-----+-----+
```

E008	Adwoa	Sarpong	Audit Review	7
E009	Kwabena	Owusu	Recruitment Drive	5
E010	Abena	Yeboah	Recruitment Drive	8
E010	Abena	Yeboah	Staff Training Program	6
E011	Yaw	Boadi	Brand Awareness Campaign	8
E012	Esi	Donkor	Brand Awareness Campaign	8
E012	Esi	Donkor	Market Research Study	6

Section 7.27

MariaDB [CaseStudy2]> create view EmpProject(empNo, projNo, totalHours) as
select w.empNo, w.projNo, SUM(hoursWorked) from Employee e, Project p, WorksOn w
where e.empNo = w.empNo and p.projNo = w.projNo group by w.empNo, w.projNo;

a)

MariaDB [CaseStudy2]> select * from EmpProject;

empNo	projNo	totalHours
E001	P01	7
E001	P02	6
E002	P01	8
E002	P02	6
E003	P02	7
E004	P03	8
E004	P04	7
E005	P03	6
E005	P04	4
E006	P04	8
E007	P05	8
E007	P06	8
E008	P05	7
E008	P06	7
E009	P07	5
E010	P07	8
E010	P08	6
E011	P09	8
E012	P09	8
E012	P10	6

b)

MariaDB [CaseStudy2]> select projNo from EmpProject where projNo = 'SCCS';
Empty set (0.004 sec)

c)

MariaDB [CaseStudy2]> select count(projNo) from EmpProject where empNo = 'E1';

count(projNo)
0

+-----+

d)

MariaDB [CaseStudy2]> select empNo, totalHours from EmpProject group by empNo;

empNo	totalHours
E001	7
E002	8
E003	7
E004	8
E005	6
E006	8
E007	8
E008	7
E009	5
E010	8
E011	8
E012	8

Section 7.28

Because the view uses DISTINCT and a single part can have multiple rows, removing a part number from the view always requires verifying the base table, while adding one never does. This asymmetry is what determines when incremental maintenance can avoid accessing Part.